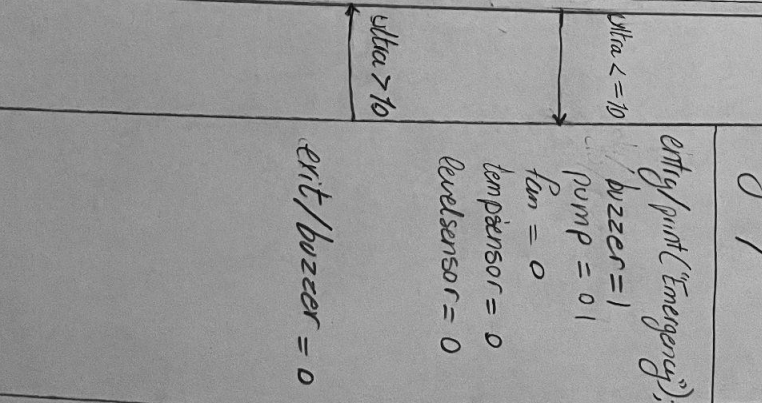
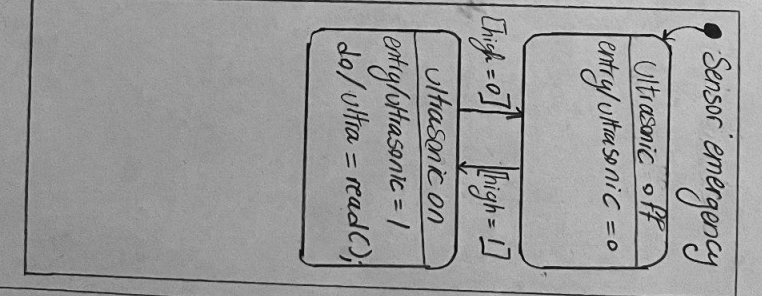
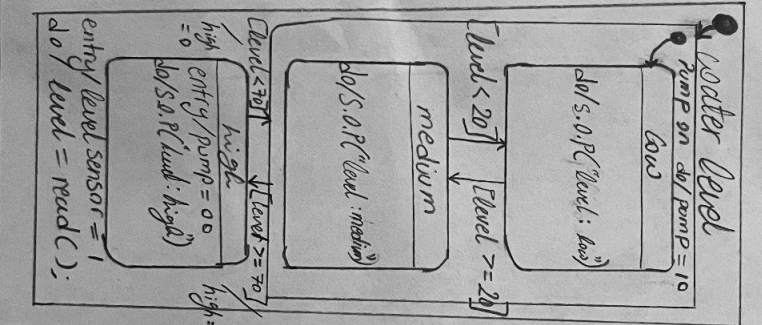
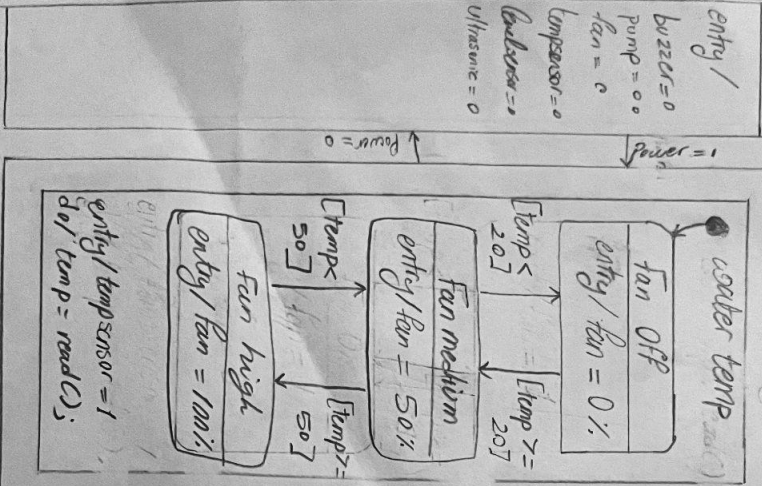


Water level monitoring

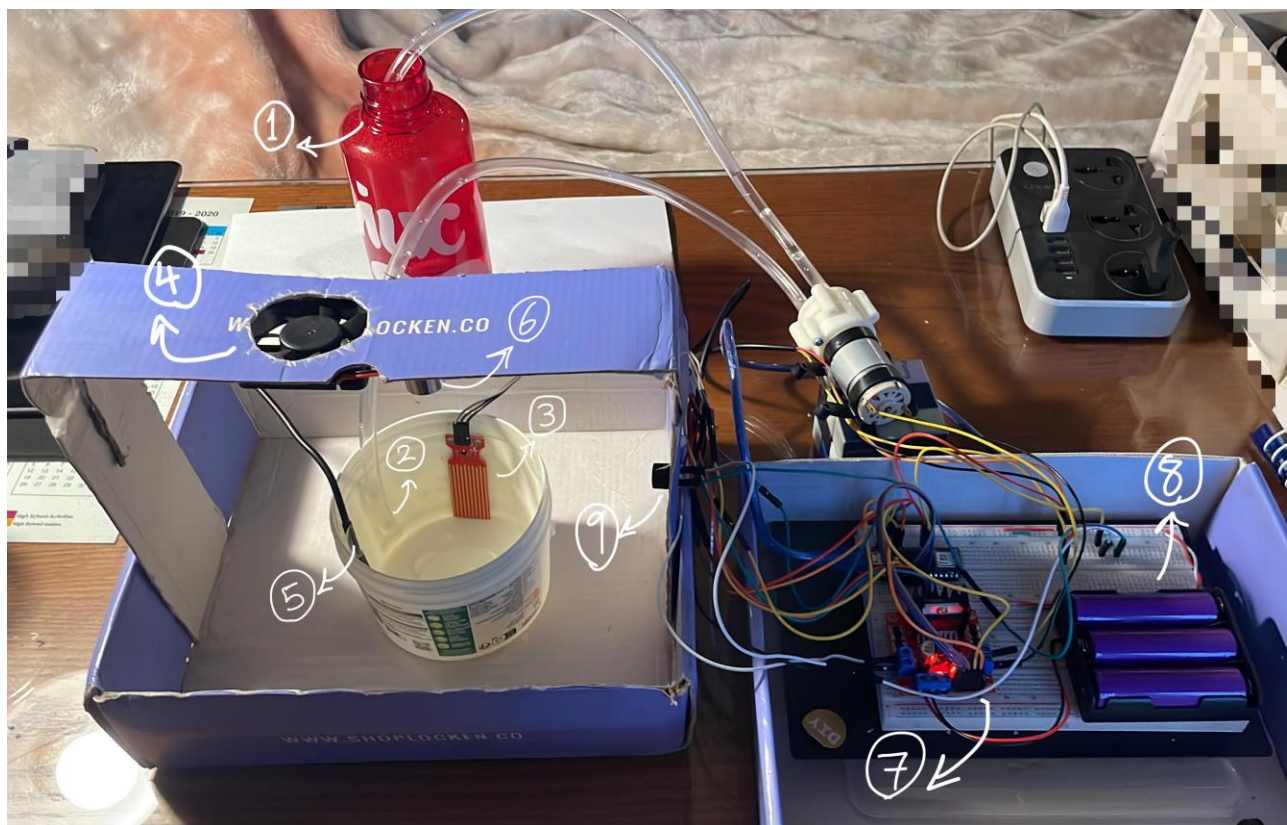
1. On - emergency

Power On



Hardware Components List with DataSheets:

- 1- water level sensor : <https://www.bitmi.ro/domains/bitmi.ro/files/files/bitmi-datasheet-senzor-nivel-masurare-apa-1794.pdf>
- 2- water temperature sensor: <https://www.analog.com/media/en/technical-documentation/data-sheets/DS18B20.pdf>
- 3- ultrasonic sensor: <https://www.handsontec.com/dataspecs/HC-SR04-Ultrasonic.pdf>
- 4- buzzer: <https://www.farnell.com/datasheets/2171929.pdf>
- 5- 2-wire fan: https://www.sameskydevices.com/product/resource/cfm-30b.pdf?srsId=AfmBOop1LhdJBy_2bcICu4OQhBdCb3LAK-AXE80lvATBYuVP743BoM
- 6- Water pump: https://www.kawamoto-global.com/web/data/ecatalog_u_pdf/2.pdf
- 7- 4.7k resistor
- 8- H-bridge: https://www.sparkfun.com/datasheets/Robotics/L298_H_Bridge.pdf
- 9- Jumpers
- 10- Breadboard: <https://cms.katranji.com/web/content/1829675>
- 11- arduino nano RP2040: <https://content.arduino.cc/assets/ABX00053-datasheet.pdf>



The purpose of this project is to monitor the water level

The pump source is marked as number 1 which is the source of our water. The destination of the pump is marked as number 2 which is where our water will be pumped. At number 3 is our water level sensor which measures the water level; if the water level is <20 then it is marked as low, if the water level is ≥ 20 and <50 then it is marked as medium, and if the water level is ≥ 50 then it is marked as high. The pump is activated when the detected water level is low or medium and is switched off when high.

At number 5 is the temperature sensor which measures the water temperature, if the water temperature is high (≥ 50 degrees celsius) the fan which is at number 4 is turned on at full speed (100%), if the temperature is medium (≥ 20 and < 50 degrees celsius), the fan will work at medium speed (50%), otherwise if the temperature is < 20 degrees celsius, the fan is be turned off.

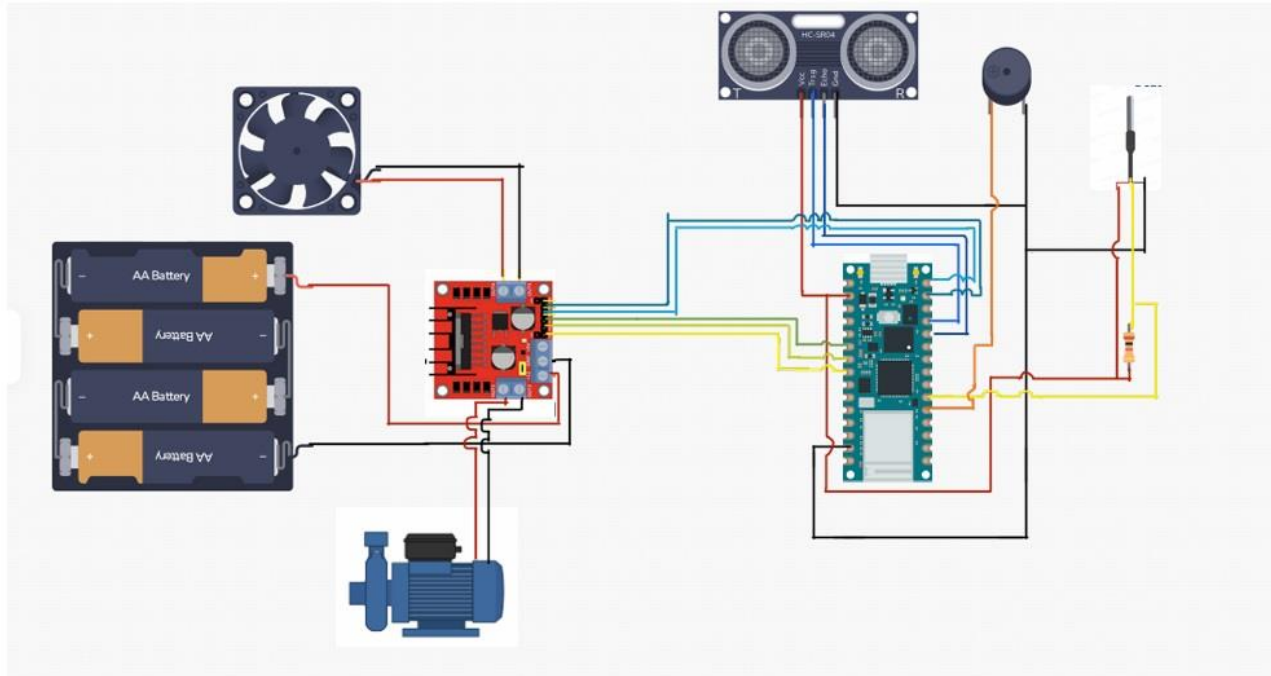
At number 6 is The ultrasonic sensor which also measures the water level and detects if the water level is too high (emergency state), so the pump's state will switch from pumping water to pumping out the water and a buzzer, which is at number 9, is turned on to alert the user about the state of emergency.

At number 7 is the h-bridge which supplies the fan and pump with 10v and splits between the 2 circuits and the Arduino to avoid an overflow.

At number 8 is the Power Source that supplies power to the h-bridge.

The whole system can be turned on and off when attaching or detaching the USB cable.

Connection Diagram



Concurrency in the System

The system uses FreeRTOS to handle concurrency. The drivers for sensors and actuators run concurrently with each other.

Concurrency Logic:

- Task Management: sensors and actuators (water level sensor and pump), (temperature sensor and fan), and (ultrasonic sensor and buzzer) are managed by a dedicated FreeRTOS tasks. These tasks run concurrently, which enables real-time monitoring and control of the system without blocking operations.
- Cooperative Scheduling: The kernel guarantees that each task has adequate CPU time so they can cooperate with each other with no preemption problems.
- Real-Time Response: With FreeRTOS, critical events can be responded to quickly, such as toggling into emergency mode or increasing/decreasing fan speeds according to temperature changes.

Github Repo Link:

https://github.com/Oraby/Embedded_Project.git